

Charlotte Wong

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EDUCATION

Tufts University

Bachelor of Science in Electrical Engineering

GPA: 3.99/4.00 (Overall); 4.00/4.00 (EE); Dean's list 3/3 semesters; Year: Sophomore

Medford, MA

August 2024 – May 2028

Relevant coursework & activities: Embedded Systems (Embedded C), Linear Algebra, Differential Equations, Physics E & M, Digital Logic (System Verilog), Circuits, Intro to Computer Science in C++, Tufts Symphony Orchestra Violin II Section Leader

EXPERIENCE

Undergraduate Research Assistant

August 2025 – Present

Massachusetts Institute of Technology: Nuclear Reactor Lab

Cambridge, MA

- Analyze data from over 100K data points collected from MIT Nuclear Reactor's sensors (DWK, Thermal Power, accelerometers, etc) with Python to characterize system behavior and variability
- Represent datasets using Pandas dataframes & model nuclear fission with iapws Python Library
- Generate mu, sigma, 3 sigma values, as well as histograms for various bin sizes for various parameters from steady state intervals

Tufts Robotics Team: Pacbot Electrical Subteam

Apr. 2025 – Present

Tufts University

Medford, MA

- Implement omnidirectional movement of a small bot via embedded C for an STM32 microcontroller to communicate with two Polulu Motor Drivers
- Write Python script to multiplex four time of flight sensors to read data from all directions alternately
- Work with small group to design PCB for STM32, motor drivers & power via KiCad software

Embedded Systems Teaching Assistant

Dec. 2025 – Present

Tufts University School of Engineering

Medford, MA

- Lead lab sections to guide groups of students through learning the peripherals & protocols of an STM32 mcu
- Grade student lab reports, homework, and exams to provide adequate feedback
- Aid students in optimizing code efficiency, working within project constraints
- Debugging via PlatformIO cross debugger (register level debugging), oscilloscopes, or circuitry debugging

Tufts Electric Racing Team: Electrical Team

September 2024 – Present

Tufts University

Medford, MA

- Soldering, cutting & crimping wires for rapid prototyping & testing on breadboards
- Basic PCB design using KiCad & Altium
- Simulate circuits with LTSpice, and test circuits using oscilloscopes & signal generators

Women in Tech Conference Board Member: Design Committee

Mar. 2025 – Oct. 2025

Tufts University

Medford, MA

- Delegate design tasks to team of 7 members; communicate with larger conference team of 30 members on goals & deadlines
- Use HTML, CSS & Wix in conjunction to design the conference website shown to sponsors and prospective attendees
- Design graphics with Figma and Procreate to advertise the conference event to on & off campus audiences

PROJECTS

Remote Controlled Car | C, PWM, UART, Flask, Socket

May 2025 - Present

- Implement an STM32 microcontroller's peripherals to drive a small car
- Use Polulu DC motor driver through GPIO and PWM
- Use UART communication to send signals to drive car wirelessly
- Design PCB for circuitry between STM32 & motor drivers via KiCad

FPGA System Verilog Snake Game | SystemVerilog, ROM, RAM, UPduino, Breadboard Prototyping

Oct. 2025 - Dec. 2025

- Synthesize ROM to store each pixel to display a unique design on VGA screen
- Implement RAM to store game frames that update at a frequency slow enough for VGA to render
- Intake serial data from NES controller to implement user input for snake movement

Electric Piano | C, breadboard prototyping, Python, Git, DAC, GPIO

April 2025 – May 2025

- Prototyped breadboard hardware by implementing buttons for each key, and additional buttons for game functions
- Displayed pressed notes and music notes on a liquid crystal display to enhance user interaction experience
- Applied digital-to-analog conversion for tone generation

TECHNICAL SKILLS

Hardware/EE tools: Oscilloscope, Altium/KiCad, I2C, DAC, ADC, Interrupts, Embedded C, STM32, PWM, GPIO

CS Languages: Java, Python, C/C++, ARM Assembly, HTML/CSS, MatLab, SystemVerilog

Libraries: pandas, NumPy, Matplotlib, Scipi

Languages: Mandarin Chinese (Native), English (Native), Spanish (Working Proficiency)